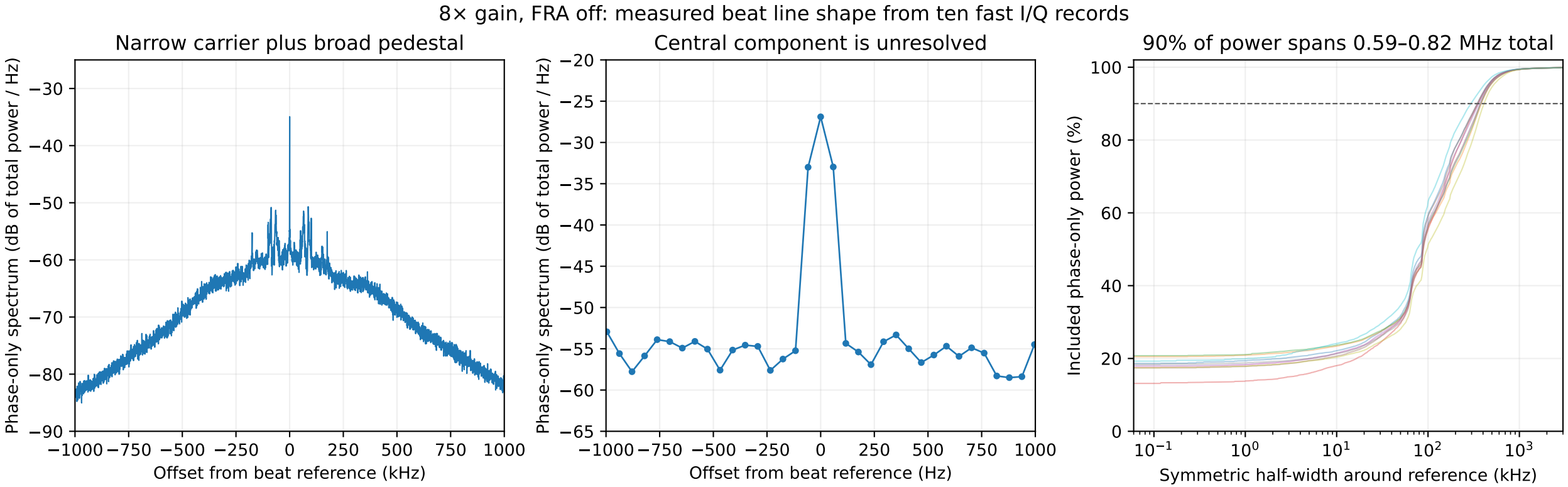

8× gain, FRA off: measured beat line shape from ten fast I/Q records
Narrow carrier plus broad pedestal
Phase-only spectrum (dB of total power / Hz)
−30
−40
−50
−60
−70
−80
−90
−1000 −750 −500 −250 0 250 500 750 1000
Offset from beat reference (kHz)
Central component is unresolved
Phase-only spectrum (dB of total power / Hz)
−20
−25
−30
−35
−40
−45
−50
−55
−60
−65
−1000 −750 −500 −250 0 250 500 750 1000
Offset from beat reference (Hz)
90% of power spans 0.59–0.82 MHz total
Included phase-only power (%)
100
80
60
40
20
0
10^{-1} 10^{0} 10^{1} 10^{2} 10^{3}
Symmetric half-width around reference (kHz)
17.07 ms per record; 58.59 Hz FFT bins; Hann ENBW 87.89 Hz for spectral plots. Finite-record coherent fraction: 13–21%.
Beat measurement includes reference-laser noise; no absolute optical linewidth inferred.